
Table of Contents

EAS 4300 HW #7 Question 2	1
Given Parameters	1
Preallocate	2
Inlet / Freestream	2
Turbojet Cycle With and Without Afterburner	3
Optimum rc	5
Figure 1: Specific Thrust vs. rc	5
Figure 2: TSFC vs. rc	6
Figure 3: Efficiencies vs. rc	7
Figure 4: Nozzle Area Ratio vs. rc	8
Save Data to CSV	9
Helper Functions	10

EAS 4300 HW #7 Question 2

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Adding an afterburner to question 1. The maximum stagnation temperature downstream of the afterburner is 2000 K. The afterburner combustion efficiency η_{ab} is 0.95 and the total pressure ratio r_{ab} is 0.97. All other efficiencies remain the same. Use a γ value of 1.4 up to the primary burner, and a value of 1.3 for the rest of the engine. Assume R is 0.287 kJ/kg·K throughout the engine. Plot the specific thrust, TSFC, η_{th} , η_p , and η_o as a function of rc , the total pressure ratio across the compressor. Consider a range of rc from 2 to 60. Assume the exhaust is ideally expanded. Also plot the nozzle area ratio as a function of rc . Comment on the changes that occur due to the addition of the afterburner (i.e., compare the plots to that of problem 1). Note that the overall fuel-to-air ratio cannot exceed the stoichiometric value ($fb + fab \leq fst$).

```
clc; clear; close all;
```

Given Parameters

```
M                = 1.8;
T04_max          = 1500;      % [K], max turbine inlet temperature
T06_ab_max       = 2000;      % [K], max afterburner stagnation temperature
delta_h_c        = 43124;     % [kJ/kg], fuel lower heating value
eta_d            = 0.9;       % diffuser efficiency
eta_c            = 0.9;       % compressor efficiency
eta_b            = 0.98;      % main burner efficiency
eta_t            = 0.92;      % turbine efficiency
eta_n            = 0.98;      % nozzle efficiency
eta_ab           = 0.95;      % afterburner combustion efficiency
rb               = 0.97;      % burner pressure ratio (P04 / P03)
rab              = 0.97;      % afterburner pressure ratio (P06 / P05)
f_st             = 0.06;      % stoichiometric fuel-to-air ratio
k                = [1.4, 1.3]; % gamma values: 1.4 (up to burner), 1.3 (rest)
R                = 0.287;     % [kJ/kg·K], gas constant
Ta              = 216.65;     % [K], ambient temperature at 15 km
```

```
Pa          = 1.2112e4;    % [Pa], ambient pressure at 15 km
delta       = 100;        % number of data points
```

Preallocate

```
rc          = linspace(2, 60, delta)'; % compressor pressure ratio range
n           = length(rc);

% Non-afterburner arrays
f_b         = zeros(n,1);
T04         = zeros(n,1);
T05         = zeros(n,1);
P05         = zeros(n,1);
I           = zeros(n,1);
TSFC        = zeros(n,1);
eta_th      = zeros(n,1);
eta_p       = zeros(n,1);
eta_0       = zeros(n,1);
Ratio       = zeros(n,1);

% Afterburner arrays
f_ab        = zeros(n,1);
f           = zeros(n,1); % total fuel (main + afterburner)
T06_ab      = zeros(n,1);
I_ab        = zeros(n,1);
TSFC_ab     = zeros(n,1);
eta_p_ab    = zeros(n,1);
eta_th_ab   = zeros(n,1);
eta_0_ab    = zeros(n,1);
Ratio_ab    = zeros(n,1);
```

Inlet / Freestream

Stagnation temperature and flight velocity:

$$T_{0a} = T_a \left(1 + \frac{\gamma_1 - 1}{2} M^2 \right), \quad u = M \sqrt{\gamma_1 R T_a}$$

The diffuser is adiabatic, so $T_{02} = T_{0a}$. Its efficiency sets the pressure recovery:

$$T_{02s} = T_a + \eta_d (T_{0a} - T_a), \quad P_{02} = P_a \left(\frac{T_{02s}}{T_a} \right)^{\gamma_1/(\gamma_1 - 1)}$$

Specific heats: $c_{p1} = \frac{\gamma_1}{\gamma_1 - 1} R$ for air ($\gamma_1 = 1.4$) and $c_{p2} = \frac{\gamma_2}{\gamma_2 - 1} R$ for burned gas ($\gamma_2 = 1.3$).

```
T0a = Ta * (1 + ((k(1) - 1)/2) * M^2);
u   = M * sqrt(R * 1000 * k(1) * Ta);
```

```
% Ideal diffuser (for T02s), then real P02
T02s = eta_d * (T0a - Ta) + Ta;
P02  = Pa * (T02s / Ta)^(k(1)/(k(1) - 1));
```

% Specific heats [kJ/kg·K]: air ($k = 1.4$) and burned gas ($k = 1.3$)
 $C_p = (k / (k - 1)) * R$;

Turbojet Cycle With and Without Afterburner

Compressor (efficiency η_c):

$$T_{03s} = T_{02} r_c^{(\gamma_1-1)/\gamma_1}, \quad T_{03} = T_{02} + \frac{T_{03s} - T_{02}}{\eta_c}, \quad w_c = c_{p1}(T_{03} - T_{02})$$

Main burner energy balance, with the stoichiometric limit $f_b \leq f_{st}$:

$$c_{p1}T_{03} + f_b \eta_b \Delta h_c = (1 + f_b) c_{p2}T_{04} \Rightarrow f_b = \frac{c_{p2}T_{04} - c_{p1}T_{03}}{\eta_b \Delta h_c - c_{p2}T_{04}}, \quad P_{04} = r_b r_c P_{02}$$

If f_b would exceed f_{st} , the burner runs at f_{st} and T_{04} is solved from the same balance.

Turbine work drives the compressor (efficiency η_t):

$$(1 + f_b) c_{p2}(T_{04} - T_{05}) = w_c, \quad T_{05s} = T_{04} - \frac{T_{04} - T_{05}}{\eta_t}$$

$$P_{05} = P_{04} \left(\frac{T_{05s}}{T_{04}} \right)^{\gamma_2/(\gamma_2-1)}$$

Afterburner energy balance, reheating the turbine exhaust to T_{06} :

$$(1 + f_b) c_{p2}T_{05} + f_{ab} \eta_{ab} \Delta h_c = (1 + f_b + f_{ab}) c_{p2}T_{06}$$

$$f_{ab} = \frac{(1 + f_b)(T_{06} - T_{05})}{\eta_{ab} \Delta h_c / c_{p2} - T_{06}}, \quad P_{06} = r_{ab} P_{05}$$

If $f_b + f_{ab}$ would exceed f_{st} , the afterburner burns only $f_{ab} = f_{st} - f_b$ and T_{06} is solved from the same balance.

Nozzle (function nozzle_exit) expands ideally to P_a from $(T_0, P_0) = (T_{05}, P_{05})$ without the afterburner, or (T_{06}, P_{06}) with it:

$$T_{es} = T_0 \left(\frac{P_a}{P_0} \right)^{(\gamma_2-1)/\gamma_2}, \quad T_e = T_0 - \eta_n(T_0 - T_{es})$$

$$M_e = \sqrt{\frac{2}{\gamma_2-1} \left(\frac{T_0}{T_e} - 1 \right)}, \quad u_e = M_e \sqrt{\gamma_2 R T_e}$$

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma_2+1} \left(1 + \frac{\gamma_2-1}{2} M_e^2 \right) \right]^{\frac{\gamma_2+1}{2(\gamma_2-1)}}$$

Performance, with $f = f_b$ without the afterburner and $f = f_b + f_{ab}$ with it:

$$I = (1 + f)u_e - u, \quad \text{TSFC} = \frac{f}{I}$$

$$\eta_{th} = \frac{(1 + f)u_e^2 - u^2}{2f\Delta h_c}, \quad \eta_p = \frac{Iu}{\frac{1}{2}[(1 + f)u_e^2 - u^2]}, \quad \eta_0 = \eta_{th}\eta_p$$

```

for j = 1:n
    % Compressor
    T03s = T0a * rc(j)^((k(1) - 1)/k(1));
    T03 = ((T03s - T0a) / eta_c) + T0a;
    W = Cp(1) * (T03 - T0a);

    % Main burner energy balance: Cp(1)*T03 + f*eta_b*dhc = (1 + f)*Cp(2)*T04
    f_b(j) = (Cp(2)*T04_max - Cp(1)*T03) / (eta_b*delta_h_c - Cp(2)*T04_max);
    T04(j) = T04_max;
    if f_b(j) > f_st
        % Stoichiometric limit: burn at f_st and find the lower T04
        f_b(j) = f_st;
        T04(j) = (Cp(1)*T03 + f_st*eta_b*delta_h_c) / ((1 + f_st)*Cp(2));
    end
    P04 = rb * rc(j) * P02;

    % Turbine exit (T05 = T06 = T07 without the afterburner)
    T05(j) = T04(j) - (W/(Cp(2)*(1 + f_b(j))));
    T05s = ((T05(j) - T04(j))/eta_t) + T04(j);
    P05(j) = P04 * (T05s/T04(j))^(k(2)/(k(2) - 1));

    % ---- Without afterburner: nozzle expansion to ambient ----
    [Ue, Me] = nozzle_exit(T05(j), P05(j), Pa, eta_n, k(2), R);
    I(j) = (1 + f_b(j))*Ue - u;
    TSFC(j) = f_b(j) / I(j);
    eta_p(j) = (I(j)*u) / ((1 + f_b(j))*(Ue^2)/2 - (u^2)/2);
    eta_th(j) = ((1 + f_b(j))*Ue^2 - u^2)/(2*f_b(j)*delta_h_c*1000);
    eta_0(j) = eta_th(j) * eta_p(j);
    Ratio(j) = area_ratio(Me, k(2));

    % ---- With afterburner ----
    % Energy balance: (1 + f_b)*Cp(2)*T05 + f_ab*eta_ab*dhc = (1 + f_b +
f_ab)*Cp(2)*T06
    f_ab(j) = ((1 + f_b(j))*(T06_ab_max - T05(j))) / ...
        (((eta_ab*delta_h_c)/Cp(2)) - T06_ab_max);
    T06_ab(j) = T06_ab_max;
    if f_b(j) + f_ab(j) > f_st
        % Overall fuel is capped at f_st: burn the rest and find the lower T06
        f_ab(j) = f_st - f_b(j);
        T06_ab(j) = ((1 + f_b(j))*T05(j) + f_ab(j)*eta_ab*delta_h_c/Cp(2)) /
(1 + f_st);
    end
    f(j) = f_b(j) + f_ab(j);

    % Afterburner exit and nozzle expansion to ambient
    P06_ab = rab * P05(j);

```

```

[Ue_ab, Me_ab] = nozzle_exit(T06_ab(j), P06_ab, Pa, eta_n, k(2), R);
I_ab(j)        = (1 + f(j))*Ue_ab - u;
TSFC_ab(j)     = f(j) / I_ab(j);
eta_th_ab(j)   = ((1 + f(j))*Ue_ab^2 - u^2) / (2*f(j)*delta_h_c*1000);
eta_p_ab(j)    = (I_ab(j)*u) / ((1 + f(j))*(Ue_ab^2)/2 - (u^2)/2);
eta_0_ab(j)    = eta_th_ab(j)*eta_p_ab(j);
Ratio_ab(j)    = area_ratio(Me_ab, k(2));

```

end

Optimum rc

```

[I_max, iI]      = max(I);
[TSFC_min, iT]   = min(TSFC);
[I_ab_max, iIab] = max(I_ab);
[TSFC_ab_min, iTab] = min(TSFC_ab);
fprintf('Without afterburner: max I = %.1f m/s at rc = %.2f; min TSFC = %.4e s/m at rc = %.2f\n', ...
        I_max, rc(iI), TSFC_min, rc(iT));
fprintf('With afterburner:      max I = %.1f m/s at rc = %.2f; min TSFC = %.4e s/m at rc = %.2f\n', ...
        I_ab_max, rc(iIab), TSFC_ab_min, rc(iTab));
fprintf('Largest total fuel-to-air ratio: %.4f (f_st = %.2f)\n', max(f), f_st);

```

Without afterburner: max I = 708.2 m/s at rc = 6.10; min TSFC = 3.5901e-05 s/m at rc = 43.01

With afterburner: max I = 1106.1 m/s at rc = 21.33; min TSFC = 4.9295e-05 s/m at rc = 20.75

Largest total fuel-to-air ratio: 0.0548 (f_st = 0.06)

Figure 1: Specific Thrust vs. rc

```

figure('Position', [100, 100, 600, 400]);
plot(rc, I, 'DisplayName', 'Non-afterburner', 'LineWidth', 1.5);
hold on
plot(rc, I_ab, '--', 'Color', 'b', 'DisplayName', 'Afterburner', 'LineWidth', 1.5);
hold off
ylim([300 1200]);
legend('Location', 'southwest');
xlabel('r_c [dim]', 'FontWeight', 'bold');
ylabel('I [m/s]', 'FontWeight', 'bold');
title('Specific Thrust vs. r_c');
grid on;

```

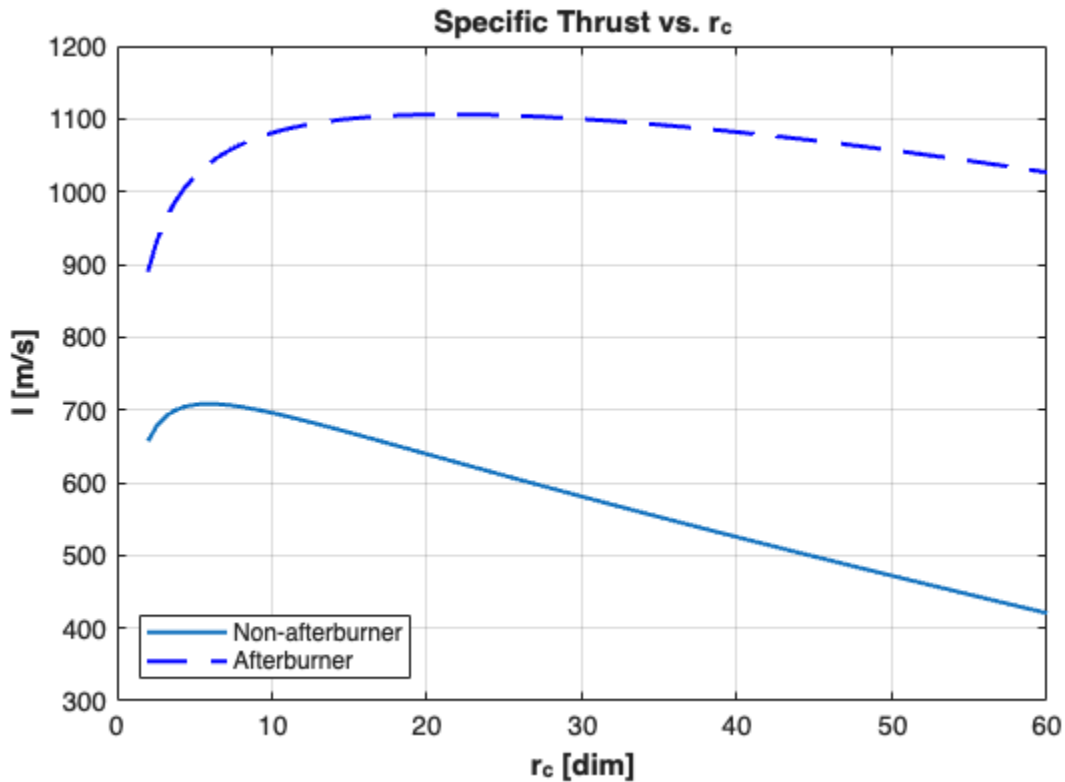


Figure 2: TSFC vs. r_c

```
figure('Position', [120, 120, 600, 400]);
plot(rc, TSFC, 'DisplayName', 'Non-afterburner', 'LineWidth', 1.5);
hold on
plot(rc, TSFC_ab, '--',
'Color', 'b', 'DisplayName', 'Afterburner', 'LineWidth', 1.5);
hold off
legend('Location', 'northeast');
ax = gca; ax.YAxis.Exponent = 0; % no scientific notation
xlabel('r_c [dim]', 'FontWeight', 'bold');
ylabel('TSFC [s/m]', 'FontWeight', 'bold');
title('TSFC vs. r_c');
grid on;
```

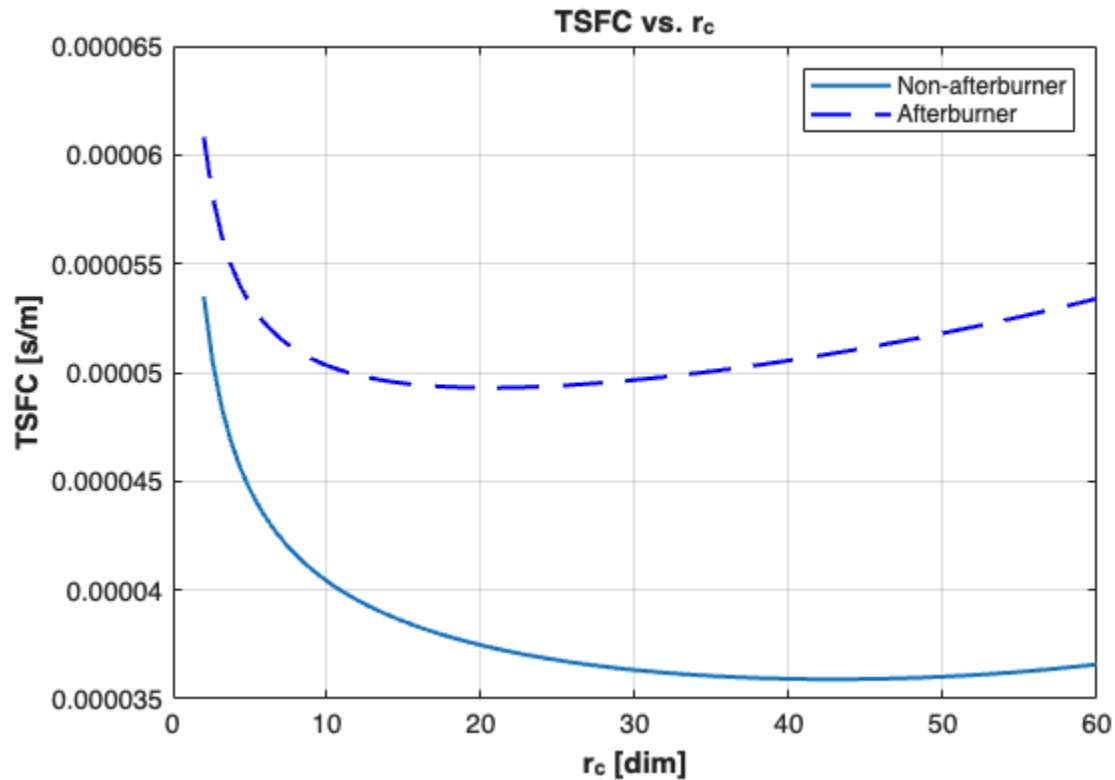


Figure 3: Efficiencies vs. r_c

```
figure('Position', [140, 140, 600, 400]);
plot(rc, eta_th, 'Color', 'b', 'LineWidth', 1.5, 'DisplayName', '\eta_{th} Non-AB');
hold on
plot(rc, eta_0, 'Color', 'k', 'LineWidth', 1.5, 'DisplayName', '\eta_{o} Non-AB');
plot(rc, eta_p, 'Color', 'r', 'LineWidth', 1.5, 'DisplayName', '\eta_{p} Non-AB');
plot(rc, eta_th_ab, '--', 'Color', 'b', 'LineWidth', 1.5, 'DisplayName', '\eta_{th} AB');
plot(rc, eta_0_ab, '--', 'Color', 'k', 'LineWidth', 1.5, 'DisplayName', '\eta_{o} AB');
plot(rc, eta_p_ab, '--', 'Color', 'r', 'LineWidth', 1.5, 'DisplayName', '\eta_{p} AB');
hold off
legend('Location', 'northwest', 'FontSize', 9);
xlabel('r_c [dim]', 'FontWeight', 'bold');
ylabel('Efficiency', 'FontWeight', 'bold');
title('Efficiencies vs. r_c');
grid on;
```

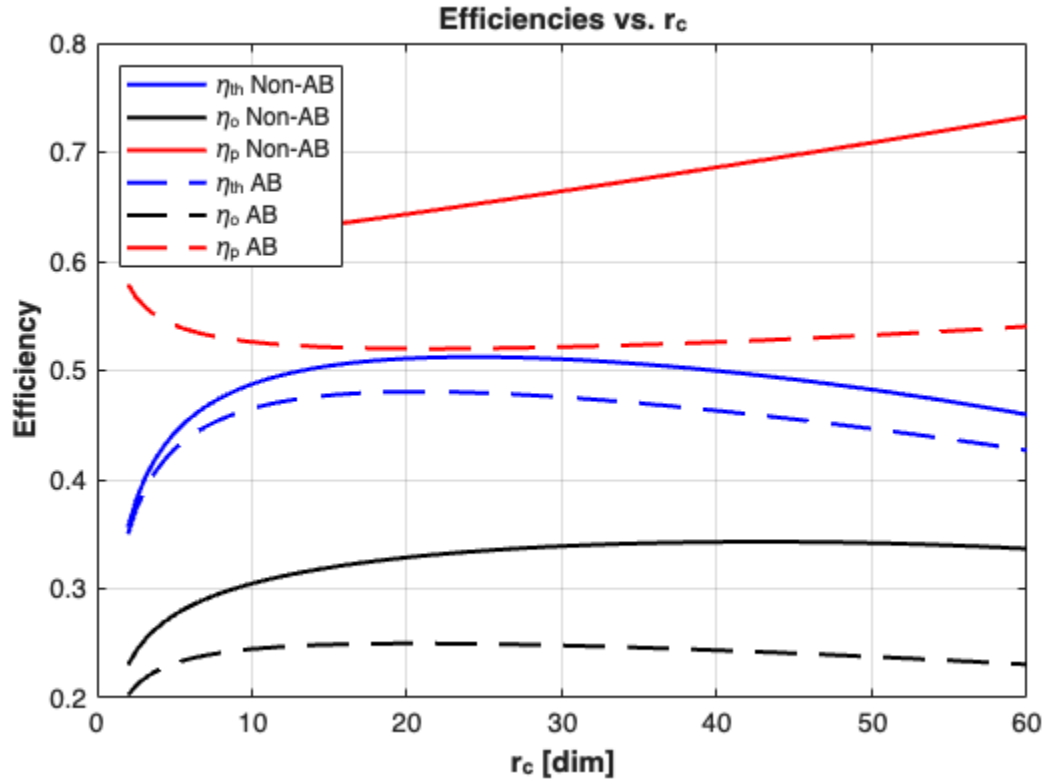
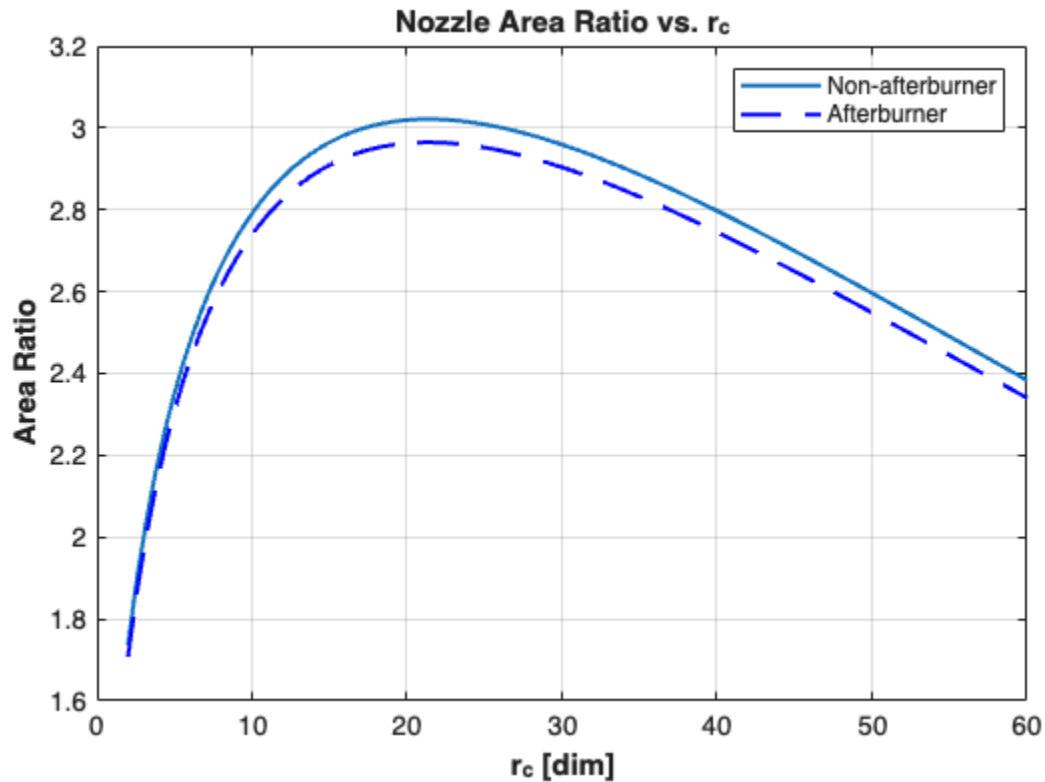


Figure 4: Nozzle Area Ratio vs. r_c

```
figure('Position', [160, 160, 600, 400]);
plot(rc, Ratio, 'DisplayName', 'Non-afterburner', 'LineWidth', 1.5);
hold on
plot(rc, Ratio_ab,
'--', 'Color', 'b', 'DisplayName', 'Afterburner', 'LineWidth', 1.5);
hold off
legend('Location', 'northeast');
xlabel('r_c [dim]', 'FontWeight', 'bold');
ylabel('Area Ratio', 'FontWeight', 'bold');
title('Nozzle Area Ratio vs. r_c');
grid on;
```



Save Data to CSV

```
data = table( ...
    rc, ...
    f_b,      f_ab,      f,      T06_ab, ...
    I,        I_ab, ...
    TSFC,     TSFC_ab, ...
    eta_th,   eta_th_ab, ...
    eta_p,    eta_p_ab, ...
    eta_0,    eta_0_ab, ...
    Ratio,    Ratio_ab, ...
    'VariableNames', { ...
    'rc', ...
    'f_b_nonAB', 'f_ab_AB', 'f_total', 'T06_AB', ...
    'I_nonAB', 'I_AB', ...
    'TSFC_nonAB', 'TSFC_AB', ...
    'eta_th_nonAB', 'eta_th_AB', ...
    'eta_p_nonAB', 'eta_p_AB', ...
    'eta_0_nonAB', 'eta_0_AB', ...
    'AR_nonAB', 'AR_AB'}});
```

```
writetable(data, 'HW7_Q2_Data.csv');
disp('Results saved to HW7_Q2_Data.csv');
```

Results saved to HW7_Q2_Data.csv

Helper Functions

```
function [Ue, Me] = nozzle_exit(T0, P0, Pa, eta_n, k, R)
    % Expand from stagnation state (T0, P0) to ambient pressure Pa
    Ts = T0 / (P0/Pa)^((k - 1)/k);      % Isentropic exit temperature
    Te = eta_n * (Ts - T0) + T0;        % Actual exit temperature
    Me = sqrt(((T0/Te) - 1)/((k - 1)/2));
    Ue = Me * sqrt(k*R*1000*Te);        % Exit velocity [m/s]
end

function AR = area_ratio(Me, k)
    % Nozzle exit-to-throat area ratio for exit Mach number Me
    AR = (1/Me) * ((2/(k+1)) * (1 + ((k-1)/2)*Me^2))^((k+1)/(2*(k-1)));
end
```

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